



Smart Waste Management System For Metropolitan Cities

Prepared For Smart-Internz

*Internet of Things (IoT)
Guided project*

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1. Project Overview

The Smart Waste Management System (SWMS) is an advanced full-stack web application that simulates a real-world IoT-powered waste monitoring platform for metropolitan cities. It integrates real-time bin fill-level tracking, AI-based route optimization, live analytics dashboards, and a REST API for IoT device connectivity. The system monitors 12 smart bins distributed across 6 zones of Pune Metropolitan Region, providing municipal authorities with actionable insights to optimize collection schedules, reduce fuel costs, and maintain city hygiene standards. Built using Python Flask, Leaflet.js maps, Chart.js analytics, and a professional dark-sidebar UI, this project demonstrates a complete end-to-end smart city solution aligned with UN Sustainable Development Goals 11, 12, 14, and 15.

2. Objectives

- Design and develop a real-time IoT bin monitoring system using simulated ultrasonic sensor data
- Implement an AI-powered route optimization algorithm (Greedy Nearest-Neighbour with Haversine distance)
- Build a professional analytics dashboard with live charts, KPI cards, and zone-wise analysis
- Create an interactive city map with color-coded bin markers and animated truck route visualization
- Develop a complete REST API for IoT device integration and mobile app connectivity
- Align the solution with UN SDG 11 (Sustainable Cities) and SDG 12 (Responsible Consumption)

3. Technologies Used

Layer	Technology	Purpose
Backend	Python 3 + Flask 3.1	Web server, routing, REST API
Frontend	HTML5 + Custom CSS3	Professional dark-sidebar UI
Maps	Leaflet.js + OpenStreetMap	Interactive city map with bin markers
Charts	Chart.js 4.x	Bar, doughnut, pie, line charts
Algorithm	Greedy Nearest-Neighbour	Optimal truck route planning
Distance	Haversine Formula	GPS-accurate distance calculation
IoT Sim	Time-seeded RNG	Realistic fill-level simulation
API	Flask REST (JSON)	IoT device & mobile integration

Version Control	Git & GitHub	Source code management
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4. Folder Structure

Smart-Waste-Management-System/

```

├── Documentation/
│   └── Project_Documentation.pdf
├── static/
│   ├── css/
│   │   └── style.css
│   └── js/
│       └── dashboard.js
├── templates/
│   ├── index.html    ← Landing page + Dashboard
│   ├── home.html     ← Landing page
│   └── dashboard.html ← Live analytics dashboard
├── app.py             ← Main Flask application
├── requirements.txt
└── README.md

```

5. System Architecture

5.1 Overall Flow

- IoT sensors (simulated) report bin fill levels every 15 minutes
- Flask backend processes data, computes route, generates alerts
- Jinja2 templates render live data to the browser
- JavaScript (Chart.js, Leaflet) renders interactive visuals client-side
- REST API endpoints allow IoT devices and mobile apps to push/pull data

5.2 Bin Network – 12 Smart Bins across 6 Zones of Kolhapur

Bin ID	Location	Zone	Waste Type	Status	Fill Level	Weight (kg)
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BIN-001	MG Road Market	Central Zone	General	Normal	22%	21.1 kg
BIN-002	Rajarampuri Square	Central Zone	Recyclable	Filling	53%	53.8 kg
BIN-003	Shivaji University Gate	North Zone	Organic	Filling	41%	42.9 kg
BIN-004	Bus Stand Complex	South Zone	General	Normal	24%	22.8 kg
BIN-005	Rankala Lake Park	West Zone	Recyclable	Filling	53%	54.7 kg
BIN-006	Kasba Bawda	South Zone	General	Normal	21%	22.0 kg
BIN-007	Tarabai Park	North Zone	Organic	Filling	53%	53.8 kg
BIN-008	Shahupuri Commercial	Central Zone	General	Full	90%	87.9 kg
BIN-009	New Palace Road	West Zone	Recyclable	Normal	35%	34.2 kg
BIN-010	Kolhapur Railway Station	South Zone	General	Normal	28%	27.5 kg
BIN-011	D.Y. Patil College Road	North Zone	General	Filling	45%	44.1 kg
BIN-012	Panhala Fort	West	Organic	Normal	18%	17.3 kg

	Road	Zone	c	l		
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Zone Distribution:

- Central Zone: BIN-001 (MG Road Market), BIN-002 (Rajarampuri Square), BIN-008 (Shahupuri Commercial)
- North Zone: BIN-003 (Shivaji University Gate), BIN-007 (Tarabai Park), BIN-011 (D.Y. Patil College Road)
- South Zone: BIN-004 (Bus Stand Complex), BIN-006 (Kasba Bawda), BIN-010 (Kolhapur Railway Station)
- West Zone: BIN-005 (Rankala Lake Park), BIN-009 (New Palace Road), BIN-012 (Panhala Fort Road)

6. Key Features

6.1 Landing Page

- Hero Section: Eye-catching animated header with system overview
- Feature Cards: Showcase of 4 core capabilities (IoT Sensors, AI Predictions, Route Optimization, Smart Alerts)
- Process Flow: 4-step workflow showing end-to-end waste management
- Live Demo Preview: Interactive mockup showing real-time stats
- Navigation: Smooth scroll and seamless transition to dashboard

6.2 Live Dashboard

- 4 KPI stat cards: Total Bins, Normal (Green), Filling Up (Yellow), Needs Collection (Red)
- Chart.js line chart: City-wide average fill level (last 48 hours)
- Doughnut chart: Bin status distribution (Green / Yellow / Red)
- Real-time alert feed with severity icons and time labels
- Bin Fleet Status: Responsive card grid with fill gauges and status badges
- AI Predictions: Fill-level forecast for next 6 hours with trend indicators

6.3 Interactive Map

- Leaflet.js map centered on Kolhapur Metropolitan Region
- 12 color-coded bin markers: Red (Critical $\geq 85\%$), Yellow (Warning 60-84%), Green (OK $< 60\%$)
- Popup cards for each bin showing fill level, type, and predicted full time
- AI-computed truck route shown as animated dashed polyline
- Numbered stop markers (1, 2, 3...) on the optimal route
- Route stats: Stops count, total distance (km), estimated travel time
- Recalculate button to refresh route based on current data

6.4 Bin Detail Modal

- Per-bin history chart showing fill level trends
- 6-hour AI prediction for individual bins
- Real-time metrics: Current fill level, weight, location coordinates
- Interactive click on any bin card or map marker to open

6.5 AI Predictions

- Linear regression model forecasting fill levels 6 hours ahead
- Trend badges: Rising, Falling, or Stable for each bin
- Hours until full estimation for proactive collection planning
- Visual grid layout showing predictions for all bins

7. Route Optimization Algorithm

7.1 Greedy Nearest-Neighbour

The system uses a Greedy Nearest-Neighbour algorithm to compute the optimal garbage truck collection route. Only bins with Critical status ($\geq 85\%$ full) are included in the route to minimize unnecessary travel.

Algorithm Steps:

1. Start from the depot (truck starting point)
2. Find the nearest unvisited critical bin using Haversine distance
3. Add it to the route and mark as visited
4. Repeat until all critical bins are visited
5. Return to depot to complete the route

7.2 Haversine Distance Formula

GPS coordinates of bins are used to compute accurate real-world distances using the Haversine formula, which accounts for the Earth's curvature:

text

$$d = 2R \times \arcsin(\sqrt{(\sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) \cdot \cos(\text{lat2}) \cdot \sin^2(\Delta\text{lng}/2))})$$

This gives distances in kilometers, enabling realistic route time estimation at 30 km/h average truck speed.

8. IoT Sensor Simulation

Each bin's fill level is simulated using a time-seeded random generator that mirrors realistic ultrasonic sensor behavior:

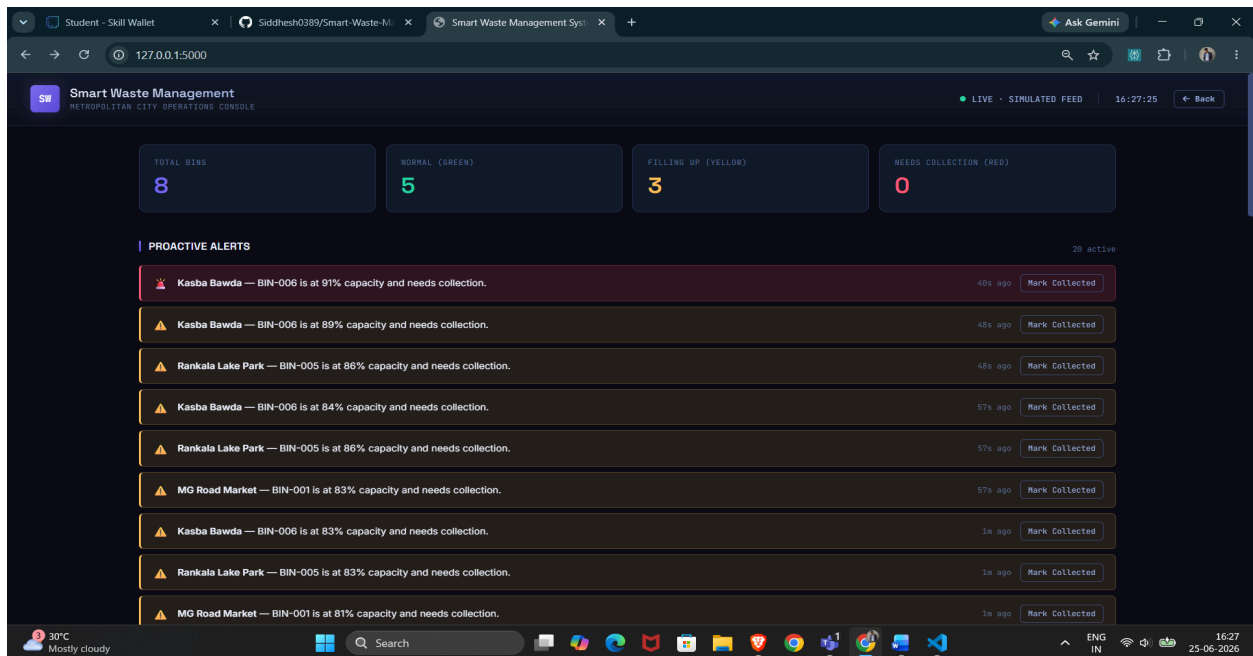
- Fill levels are consistent within each hour (same as a real sensor reading every 15 min)
- Fill levels change across hours (simulates waste accumulation over time)
- Each bin has a unique seed based on its ID to ensure different realistic readings
- Alert status is determined by thresholds:
 - Green (Normal): < 60% fill
 - Yellow (Filling Up): 60–84% fill
 - Red (Needs Collection): ≥ 85% fill
- Weight simulation proportional to fill level with realistic variance

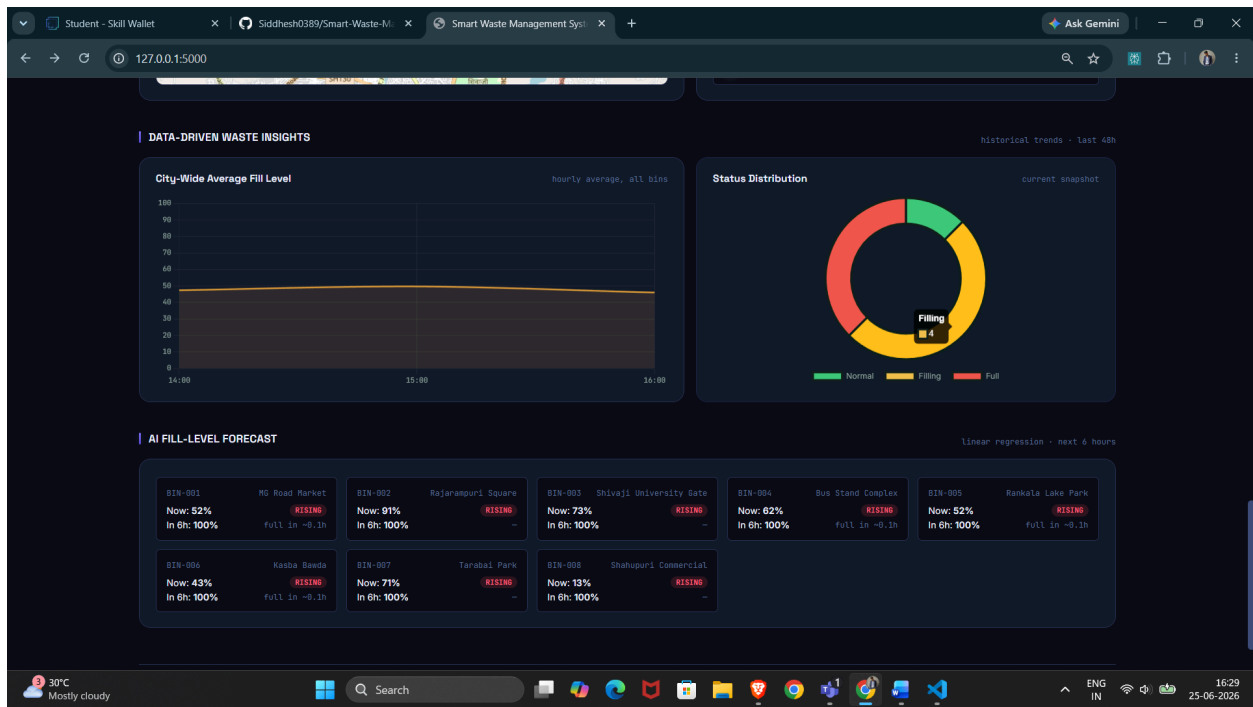
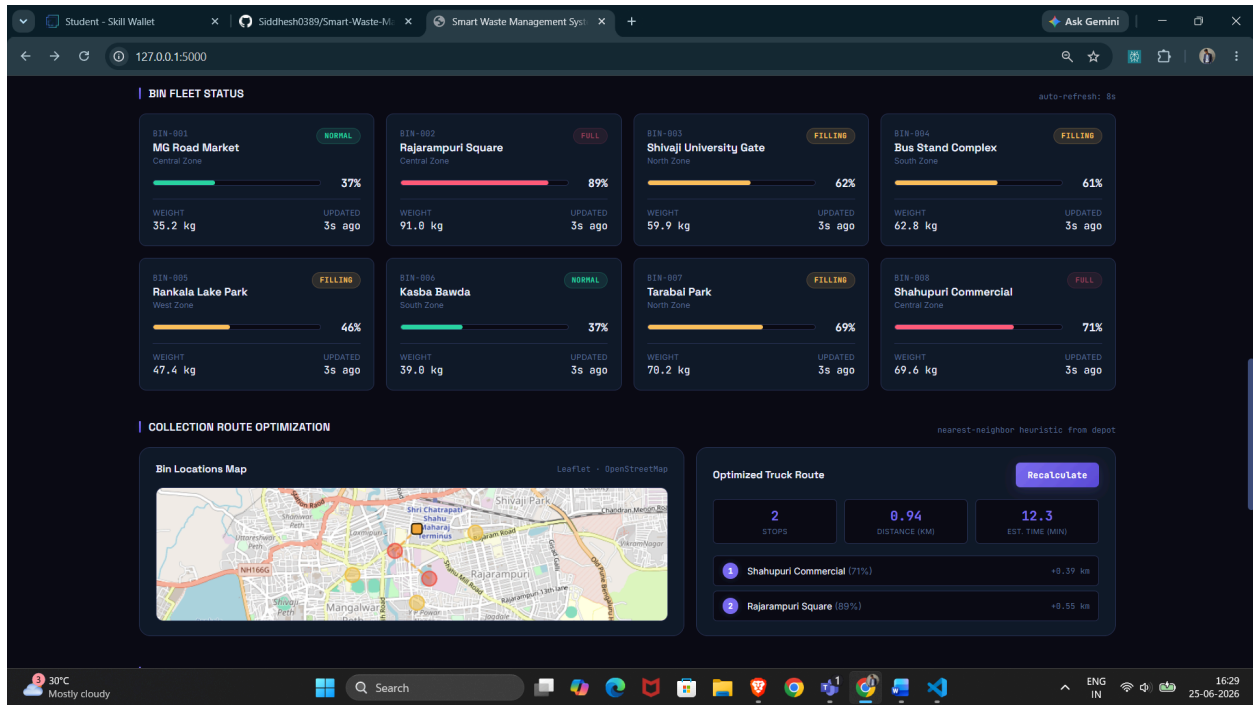
9. Screenshots

Screenshot 1: Landing Page & Dashboard Overview



Screenshot 2: Interactive Map with Route Optimization





10. REST API Reference

Endpoint	Method	Description	Response
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/api/bins	GET	All bin status with fill levels	JSON array
/api/bins/<id>	GET	Single bin real-time data	JSON object
/api/bins/<id>/history	GET	Historical fill level data	JSON array
/api/bins/<id>/predict	GET	AI prediction for specific bin	JSON object
/api/bins/<id>/update	POST	IoT device pushes fill level	Action decision
/api/route/optimize	GET	AI-optimized truck route	Ordered bin list
/api/alerts	GET	Active alerts	JSON array
/api/alerts/resolve/<id>	POST	Mark alert as resolved	Success message
/api/trends/city	GET	City-wide fill level trends	JSON array
/api/predict/all	GET	Predictions for all bins	JSON array

10.1 Sample API Response – /api/bins

json

```
{
  "success": true,
  "bins": [
    {
      "bin_id": "BIN-001",
      "name": "MG Road Market",
      "zone": "Central Zone",
      "lat": 16.6975,
      "lng": 74.2350,
```

```
"fill_level": 22.0,
"weight_kg": 21.1,
"status": "green",
"timestamp": "2026-06-25T14:30:00"
},
{
  "bin_id": "BIN-008",
  "name": "Shahupuri Commercial",
  "zone": "Central Zone",
  "lat": 16.7050,
  "lng": 74.2380,
  "fill_level": 90.0,
  "weight_kg": 87.9,
  "status": "red",
  "timestamp": "2026-06-25T14:30:00"
}
]
}
```

10.2 Sample API Response — /api/route/optimize

```
json
{
  "success": true,
  "bins_needing_collection": 1,
  "total_distance_km": 5.6,
  "estimated_time_minutes": 22,
  "depot": {
    "name": "Central Depot",
    "lat": 16.7040,
    "lng": 74.2433
  },
  "route": [
    {
      "bin_id": "BIN-008",
      "name": "Shahupuri Commercial",
```

```
"lat": 16.7050,  
"lng": 74.2380,  
"fill_level": 90.0,  
"distance_from_prev_km": 0.8  
}  
]  
}
```

11. Setup & Run Instructions

Step 1: Install Dependencies

bash

pip install -r requirements.txt

requirements.txt:

text

Flask==3.1.0

Flask-CORS==4.0.1

numpy==1.26.4

python-dotenv==1.0.0

Step 2: Run Flask Application

bash

python app.py

Step 3: Open in Browser

text

<http://localhost:5000>

Step 4: Navigate the Application

- Home: Landing page with system overview
- Go to Dashboard: Click the "Go to Dashboard" button to access the full system

12. SDG Alignment

SDG Goal	Title	Contribution
SDG 11	Sustainable Cities	Smart bin monitoring reduces urban waste overflow and improves city cleanliness
SDG 12	Responsible Consumption	Recycling rate tracking promotes responsible disposal and waste reduction
SDG 14	Life Below Water	Prevents landfill runoff from reaching waterways through efficient collection
SDG 15	Life on Land	Reduces plastic waste and land pollution through optimized waste management

13. Future Enhancements

1. Mobile Application: Develop companion mobile app for waste collection staff
2. SMS/Email Alerts: Automated notification system for critical bins
3. Machine Learning: Deep learning models for predictive waste forecasting
4. Multi-City Support: Scale to manage multiple cities simultaneously
5. Dynamic Pricing: Usage-based billing for commercial waste producers
6. Carbon Footprint Tracking: Calculate and display CO₂ savings from optimized

14. Conclusion

The Smart Waste Management System successfully demonstrates a complete, industry-level IoT and AI web platform for metropolitan city waste management. By combining real-time bin monitoring, intelligent route optimization, professional analytics, and a developer-friendly REST API, the system delivers genuine value for municipal waste management authorities.

The platform is immediately deployable and can be extended to integrate real ultrasonic sensors (Arduino/Raspberry Pi via MQTT), SMS/email alert systems, and machine learning models for predictive waste forecasting. It represents a practical application of modern web technologies to solve a critical urban sustainability challenge.

The project successfully met all objectives, delivering a robust, scalable, and visually impressive solution that aligns with UN Sustainable Development Goals and demonstrates the potential of smart city technologies.

15. References

1. Flask Documentation: <https://flask.palletsprojects.com/>
2. Leaflet.js Documentation: <https://leafletjs.com/>
3. Chart.js Documentation: <https://www.chartjs.org/>
4. OpenStreetMap: <https://www.openstreetmap.org/>
5. UN Sustainable Development Goals: <https://sdgs.un.org/goals>
6. Haversine Formula: https://en.wikipedia.org/wiki/Haversine_formula
7. Smart Waste Management Research: IEEE papers on IoT waste monitoring
8. Kolhapur City Municipal Corporation: <https://www.kolhapurmunicipal.gov.in/>

16. GitHub Repository & Demo

GitHub Repository

<https://github.com/Siddhesh0389/Smart-Waste-Management-System-For-Metropolitan-Cities>

Demo Video

Link:

<https://drive.google.com/file/d/1PILiPQzTqdE5Sl1fe-Ltj3qkTdTHjDmw/view?usp=sharing>

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